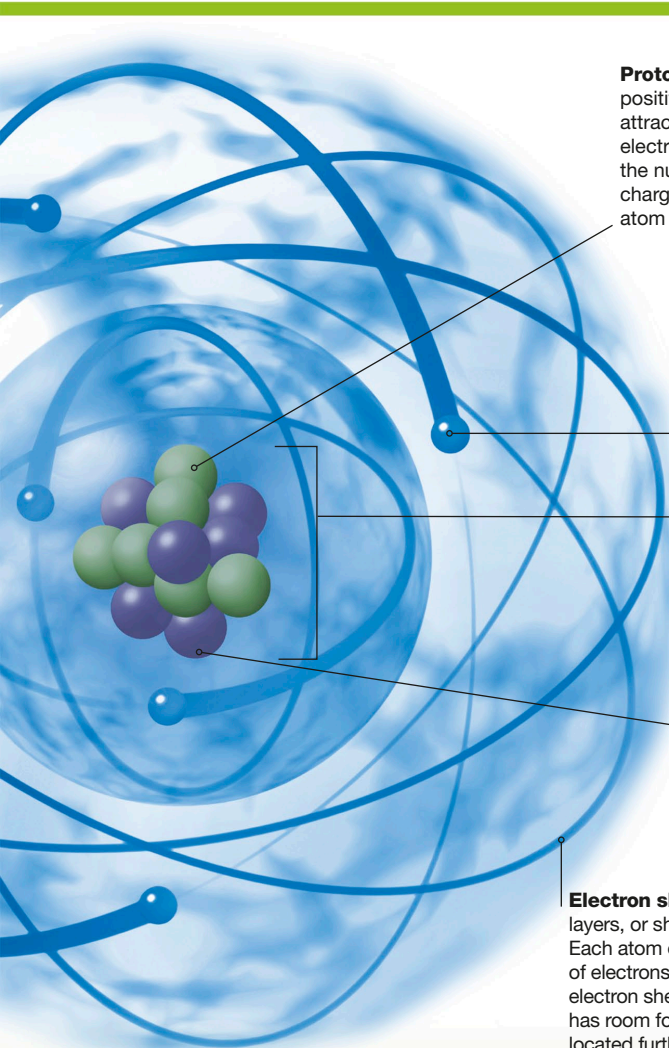




Protons: These particles have a positive electrical charge. This charge attracts the negatively charged electrons, holding them in place around the nucleus. The positive and negative charges cancel out each other, so an atom has a neutral charge.

Electrons: Negatively charged electrons were the first type of subatomic particle to be discovered. They are almost 2,000 times smaller than protons.

Nucleus: Almost all of an atom's weight is packed inside the nucleus. Even so, it is 100,000 times smaller than the atom as a whole. Most of an atom is empty space.

Neutrons: A neutron has no overall electrical charge, and helps protons cluster together in the nucleus. It adds weight to the atom.

Electron shell: Electrons are arranged in layers, or shells, around the outside of an atom. Each atom of an element has a unique number of electrons, arranged in a particular way in its electron shells. The shell closest to the nucleus has room for just two electrons, but shells located further out have room for more. Only the electrons in the outermost shell participate in chemical reactions.